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Goal

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}}_{A \in M_2(\mathbb{R})} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{x \in \mathbb{R}^2}$$

$$\Leftrightarrow \dot{x} = Ax = P \wedge \underbrace{P^{-1}x}_{z} \\ \uparrow \quad \quad \quad \downarrow \\ \text{\textcolor{red}{\Lambda}} \quad \quad \quad \text{\textcolor{red}{z}} \\ \text{\textcolor{red}{\Lambda}} \text{ Lambda}$$

Let $P^{-1}x = z$, then

$$\begin{aligned} \dot{z} &= \frac{d}{dt}(P^{-1}x) = P^{-1} \frac{dx}{dt} = P^{-1} \dot{x} = \cancel{P^{-1}(P \wedge P^{-1}x)} \\ &= \text{\textcolor{yellow}{\Lambda}} z \end{aligned}$$

i.e., $\dot{z} = \Lambda z$, i.e.,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \Leftrightarrow \dot{z}_1 = z_1 \quad \& \quad \dot{z}_2 = 5z_2$$

$$\Rightarrow z_1 = e^t z_1(0) \quad \& \quad z_2 = e^{5t} z_2(0)$$

$$\Rightarrow \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} e^t & \\ & e^{5t} \end{bmatrix} \begin{bmatrix} z_1(0) \\ z_2(0) \end{bmatrix} \quad z(t) \in \mathbb{R}^2$$

$$\Rightarrow P^{-1}x(t) = \begin{bmatrix} e^t & \\ & e^{5t} \end{bmatrix} P^{-1}x(0)$$

$$\Rightarrow x(t) = P \begin{bmatrix} e^t & 0 \\ 0 & e^{5t} \end{bmatrix} P^{-1}x(0)$$

Define the **matrix exponential** $\exp: M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$,

$$\exp(A) = e^A := \sum_{k \geq 0} \frac{A^k}{k!} = I_n + \sum_{k \geq 1} \frac{A^k}{k!}$$

then $x(t) = e^{tA} x_0 = \exp(tA) x_0$ is the sol'n to the IVP

$$\dot{x} = Ax, \quad x(0) = x_0 \quad (x \in \mathbb{R}^n, A \in M_n(\mathbb{R}))$$

By the existence & uniqueness thm, it suffices to verify that $x(t) = e^{tA} x_0$ satisfies the IVP, i.e., both the IC, and the ODE,

• At $t=0$,

$$x(0) = \exp(0A) x_0 = \exp(0_{n \times n}) x_0$$

$$= \left(I_n + \sum_{k \geq 1} \frac{0_{n \times n}^k}{k!} \right) x_0 = I_n x_0 = x_0$$

$$\bullet \frac{dx}{dt} = \frac{d}{dt} [\exp(tA) x_0] = \left(\frac{d}{dt} \exp(tA) \right) x_0$$

$$= \frac{d}{dt} \left(I_n + \sum_{k \geq 1} \frac{(tA)^k}{k!} \right) x_0$$

$$= \left(\sum_{k \geq 1} k \frac{t^{k-1} A^k}{k!} \right) x_0$$

$$= A \left(\sum_{k \geq 1} \frac{(tA)^{k-1}}{(k-1)!} \right) x_0$$

$$= A \left(\left(\sum_{k \geq 0} \frac{(tA)^k}{k!} \right) x_0 \right)$$

$$= A \left(\exp(tA) x_0 \right)$$

If A is diagonalizable, then

$$\exp(tA) = P e^{t\Lambda} P^{-1} = P \begin{bmatrix} e^{t\lambda_1} & & \\ & \ddots & \\ & & e^{t\lambda_n} \end{bmatrix} P^{-1}$$

where

$$A = P \Lambda P^{-1}, \text{ w/ } \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

$$= \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

Indeed,

$$A = P \Lambda P^{-1} \Rightarrow A^k = P \Lambda^k P^{-1} = P \begin{bmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{bmatrix} P^{-1}$$

$$\Rightarrow (tA)^k = P \underbrace{\begin{bmatrix} t^k \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & t^k \lambda_n^k \end{bmatrix}}_{(t\Lambda)^k} P^{-1}$$

$$\Rightarrow \exp(tA) = \sum_{k \geq 0} \frac{(tA)^k}{k!} = \sum_{k \geq 0} P \frac{(t\Lambda)^k}{k!} P^{-1}$$

$$= P \sum_{k \geq 0} \frac{(t\Lambda)^k}{k!} P^{-1}$$

$$= p \left(\sum_{k \geq 0} \begin{pmatrix} t^k \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & t^k \lambda_n^k \end{pmatrix} / k! \right) p^{-1}$$

$$= p \left(\sum_{k \geq 0} \begin{pmatrix} t^k \lambda_1^k / k! & & 0 \\ & \ddots & \\ 0 & & t^k \lambda_n^k / k! \end{pmatrix} \right) p^{-1}$$

$$= p \left(\begin{pmatrix} \sum_{k \geq 0} t^k \lambda_1^k / k! & & \\ & \ddots & \\ & & \sum_{k \geq 0} t^k \lambda_n^k / k! \end{pmatrix} \right) p^{-1}$$

$$= p \left(\begin{pmatrix} e^{t\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{t\lambda_n} \end{pmatrix} \right) p^{-1}$$

Given $A \in M_n(\mathbb{R})$, we can also use Laplace Transform to calculate $\exp(tA)$, which is the solution to the matrix IVP

$$\dot{\Phi} = A\Phi, \quad \Phi(0) = I_n$$

$$(\Phi(t) \in M_n(\mathbb{R}) \quad \forall t \in \mathbb{R})$$

$$(\Phi(\cdot) : \mathbb{R} \rightarrow M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}, \quad t \mapsto \Phi(t))$$

Apply Laplace Transform to the IVP, we get

$$s(\mathcal{L}\Phi)(s) - I_n = A(\mathcal{L}\Phi)(s)$$

$$\Rightarrow (sI_n - A) (\mathcal{L}\Phi)(s) = I_n$$

$$\Rightarrow (\mathcal{L}\Phi)(s) = (sI_n - A)^{-1}$$

$$\Rightarrow \Phi(t) = \mathcal{L}^{-1}\{(sI_n - A)^{-1}\}$$

Consider $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

$$(\mathcal{L}\Phi)(s) = \left[s \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} \right]^{-1}$$

$$= \begin{pmatrix} s-4 & -1 \\ -3 & s-2 \end{pmatrix}^{-1} = \frac{1}{(s-1)(s-5)} \begin{pmatrix} s-2 & +1 \\ +3 & s-4 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{s-2}{(s-1)(s-5)} & \frac{1}{(s-1)(s-5)} \\ \frac{3}{(s-1)(s-5)} & \frac{s-4}{(s-1)(s-5)} \end{pmatrix}$$

$$\Rightarrow \exp(tA) = \Phi(t) = \begin{pmatrix} \mathcal{L}^{-1}\left(\frac{s-2}{(s-1)(s-5)}\right) & \mathcal{L}^{-1}\left(\frac{1}{(s-1)(s-5)}\right) \\ \mathcal{L}^{-1}\left(\frac{3}{(s-1)(s-5)}\right) & \mathcal{L}^{-1}\left(\frac{s-4}{(s-1)(s-5)}\right) \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} e^t + 3e^{5t} & -e^t + e^{5t} \\ -3e^t + 3e^{5t} & 3e^t + e^{5t} \end{pmatrix}$$

If A is diagonalizable, then the general sol'n to the $\dot{x} = Ax$, $x(t) \in \mathbb{R}^n \forall t$ is given by

$$x(t) = c_1 e^{t\lambda_1} v_1 + c_2 e^{t\lambda_2} v_2 + \dots + c_n e^{t\lambda_n} v_n$$

where $c_1, \dots, c_n$ are constants

$$\text{and } A v_k = \lambda_k v_k, \quad k=1, \dots, n$$

For example $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

The general solution to $\dot{x} = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} x$ is given by

$$x(t) = c_1 e^t \begin{pmatrix} 1 \\ -3 \end{pmatrix} + c_2 e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

where $c_1, c_2 \in \mathbb{R}$ are arbitrary constants.

Note that $\{e^{\lambda_1 t} v_1, \dots, e^{\lambda_n t} v_n\}$ form a "fundamental set/system" of solutions to the ODE $\dot{x} = Ax$
usually we write

$$\Psi(t) = \begin{bmatrix} e^{\lambda_1 t} v_1 & \dots & e^{\lambda_n t} v_n \\ | & & | \\ | & & | \end{bmatrix}_{n \times n}$$

so the general sol'n is given by $x(t) = \Psi(t) c$
w/ $c \in \mathbb{R}^n$ is an arbitrary vector in $\mathbb{R}^n$

Rem $\exp(tA) = \Psi(t) \underbrace{\Psi^{-1}(0)}_{[\Psi(0)]^{-1}}$

$\Phi(0) = I_n$ $\Phi(t)$

Diagonalizable matrices

Special Case I All eigenvalues are distinct

Special Case II $A = A^T$ for $A \in M_n(\mathbb{R})$ (real symmetric matrices)

$$\left[\begin{array}{l} \text{or } A = \overline{A}^T = \overline{A^T} = A^* = A^H \quad \text{for } A \in M_n(\mathbb{C}) \\ \quad \quad \quad \nwarrow \quad \nearrow \\ \quad \quad \quad \text{conjugate} \\ \quad \quad \quad \text{transpose} \\ \text{We say } A \text{ is Hermitian if } A = A^H \end{array} \right]$$

Thm $A \in M_n(\mathbb{C})$ w/ distinct eigenvalues, then
 A is diagonalizable.

i.e., all eigenvectors are linearly independent

i.e.,

$$A = \begin{bmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{bmatrix}^{-1}$$

"Pf" Say $n=3$, and A has distinct eigenvalues
 $\lambda_1, \lambda_2, \lambda_3$, w/ corresponding eigenvectors v_1, v_2, v_3

$$(Av_1 = \lambda_1 v_1, Av_2 = \lambda_2 v_2, Av_3 = \lambda_3 v_3)$$

We want to show that v_1, v_2, v_3 are linearly independent, i.e.,

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \Rightarrow c_1 = c_2 = c_3 = 0 \in \mathbb{R}^1$$

Indeed, assume that $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \in \mathbb{R}^n$

then

$$(A - \lambda_2 I)(A - \lambda_3 I)(c_1 v_1 + c_2 v_2 + c_3 v_3) = 0 \in \mathbb{R}^n$$

but by distributivity

$$\begin{aligned} \text{LHS} &= \underline{c_1 (\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3) v_1} \\ &\quad + c_2 \cancel{(\lambda_2 - \lambda_2)}^0 (\lambda_2 - \lambda_3) v_2 \\ &\quad + c_3 (\lambda_3 - \lambda_2) \cancel{(\lambda_3 - \lambda_3)}^0 v_3 = \underline{0} = \text{RHS} \end{aligned}$$

$$\Rightarrow \underbrace{c_1 (\lambda_1 - \lambda_2)}_{\neq 0} \underbrace{(\lambda_1 - \lambda_3)}_{\neq 0} \underbrace{v_1}_{\neq 0} = 0 \in \mathbb{R}^n$$

$$\Rightarrow c_1 = 0$$

similar for c_2 & c_3

□